

## **Consolidated Transdermal-to-CNS Delivery Platform for Rabies Post-Exposure Therapy: MN-Iontophoresis Priming with RVG-Targeted, Dual-Antiviral Nanoparticles**

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### **Abstract**

Rabies remains almost uniformly fatal post-symptom onset due to rapid neuroinvasion and poor central nervous system (CNS) drug exposure. From the cited literature, dissolving microneedles (MNs) enhanced with iontophoresis can outperform intramuscular vaccination in immunogenicity, but do not address CNS drug levels. RVG-peptide functionalization of nanoparticles enables receptor-mediated transport across the blood–brain barrier (BBB) and significantly improves survival in neurotropic viral infection models. Yet, current systems carry a single antiviral and lack a peripheral “front-end” that accelerates dermal uptake. Ranpirnase shows strong in vitro antirabies activity but failed in vivo due to delivery/BBB limits, highlighting the need for targeted CNS exposure. Building on these findings, we propose an integrated system: a disposable HA/PVP dissolving MN patch loaded with inactivated rabies antigen for rapid skin-resident APC priming plus a co-delivered suspension of RVG-functionalized, chitosan-coated PLGA & MSN core–shell nanoparticles co-encapsulating Favipiravir (T-705) and Ranpirnase. Iontophoresis ( $\leq 0.5 \text{ mA}\cdot\text{cm}^{-2}$ ) drives nanoparticles through MN-created low-resistance microchannels to systemic circulation; RVG then targets neuronal receptors for BBB transit and CNS delivery. We model burst/sustained release to maintain peripheral and CNS drug levels above in vitro  $\text{IC}_{50}/\text{IC}_{90}$  ranges while minimising cytotoxicity. The approach aims to enhance drug response, efficacy, and targeting while mitigating the drawbacks of current methods.

## 1. Introduction

**Disease & context.** Rabies (Lyssavirus) causes progressive encephalitis with ~100% case fatality once symptomatic. Effective pre-/post-exposure prophylaxis exists, but treatment after neuroinvasion fails largely because antivirals don't reach the CNS at therapeutic levels.

The existing delivery systems demonstrate:

- **Transdermal vaccination via MNs w/w/o iontophoresis:** HA/PVP dissolving MNs with iontophoresis ("MNvi/MNdi") significantly boosted dermal delivery and immune readouts vs MN alone and vs conventional IM injection. Beagle-dog studies showed ~206% IgG increase over untreated and RVNA titers ~2.2× threshold with MN+iontophoresis; MNs themselves provided ~73% IgG increase and ~1.2× RVNA threshold. Histology confirmed epidermal grooves without deep vascular injury.
- **Polymer nanoparticles for antigen delivery:** PLGA-b-PEG and CS-PEG nanoparticles carrying whole attenuated rabies antigen showed sustained release, lymphatic targeting potential, immune stimulation in vitro, and negligible hemolytic/genotoxic effects, with further surface functionalization (BSA/casein/ovalbumin/starch) to tune charge/stealth.
- **RVG-functionalized CNS-targeted antivirals:** RVG-peptide (29 aa from rabies G protein) on mesoporous silica nanoparticles (MSNs) loaded with Favipiravir improved brain delivery and yielded 77% survival in a mouse neurotropic virus model, with ~14% drug loading and ~31% cumulative release by 192 h.
- **Ranpirnase (TMR-001) antiviral:** Potent in-vitro inhibition of rabies virus release and cell-to-cell spread ( $IC_{50}$  ~0.2–2 nM for release; 20–600 nM for spread) but no protection in hamsters at 0.1 mg·kg<sup>-1</sup> daily (various routes), likely due to inadequate local/CNS exposure and BBB transit.

### Knowledge gaps/limitations motivating a new system.

1. **CNS exposure barrier:** MN/iontophoresis vaccinates well but does not deliver antivirals to the brain.
2. **Single-modality antivirals:** RVG & MSN carried only Favipiravir; monotherapy risks resistance and may need high dosing.
3. **BBB transit of proteins:** Ranpirnase is potent but not BBB-permeant in its current form.
4. **Kinetics alignment:** Need fast peripheral suppression and sustained CNS levels; existing systems address one but not both phases.

**Hypothesis.** A **two-stage, single-procedure platform**-MN-iontophoresis front-end for rapid transdermal entry + RVG-targeted, dual-antiviral nanoparticles (Favipiravir + Ranpirnase) with staged release-will (i) accelerate peripheral viral suppression, (ii) deliver antivirals into the CNS at therapeutic levels, and (iii) enhance protective immunity, thereby improving overall efficacy and safety compared with current single-method approaches.

## 2. Specific Aims

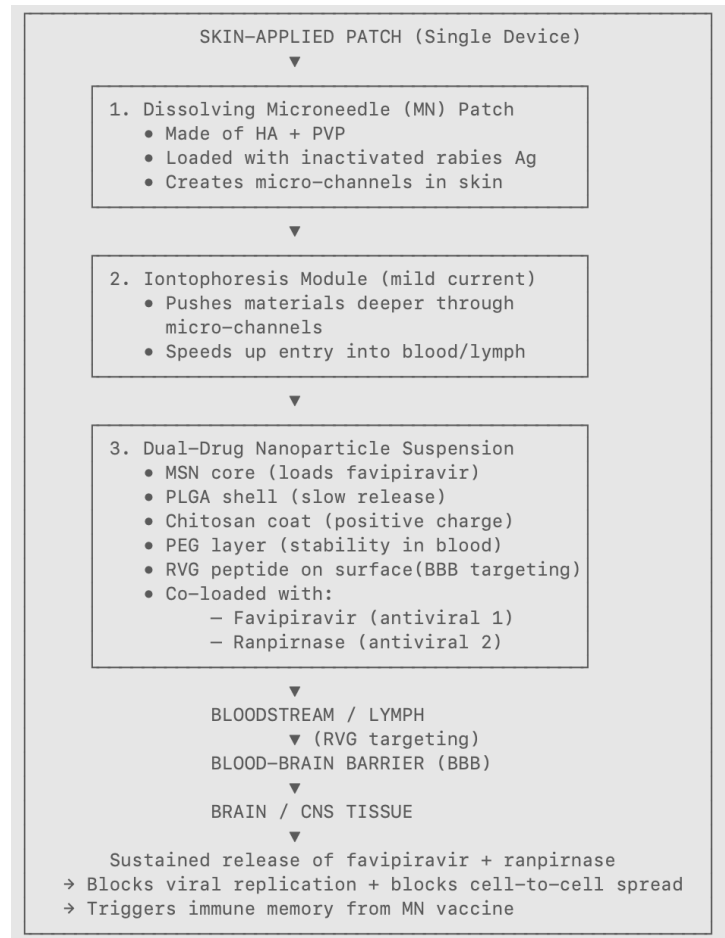
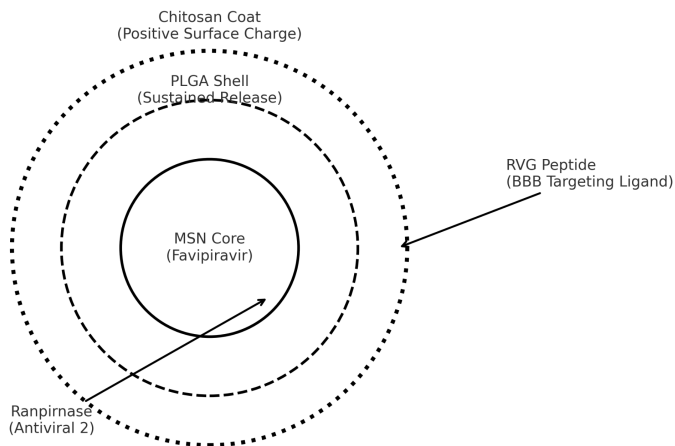
1. **Design criteria.** Define materials/architecture for a skin-to-brain delivery train that is minimally invasive, immuno-enhancing, and BBB-targeting. (MNs + iontophoresis; RVG-functionalized core-shell nanoparticles).
2. **Formulation & targeting.** Propose a PLGA-shell & MSN core nanoparticle, CS surface coat, RVG ligand, co-loading Favipiravir + Ranpirnase, optimized for charge, size, and release.
3. **Transport & release modeling.** Predict dermal flux (MNs  $\pm$  iontophoresis) and CNS delivery kinetics to maintain concentrations above reported in-vitro  $IC_{50}/IC_{90}$  of payloads while minimizing cytotoxicity.
4. **Expected performance & risks.** Benchmark immunogenicity, antiviral effect, safety, and practical barriers vs each source method, and outline mitigation.

## 3. Methods and Results (Design + Calculations)

### 3.1 System Architecture

- **Stage-1 (skin entry & priming):** A dissolving **HA/PVP** MN patch (needle height 300–500  $\mu\text{m}$  per attached work) with **PEG-400** plasticizer, loaded in backing with inactivated rabies antigen for dermal APC activation-already shown to outperform IM in IgG/RVNA with iontophoresis. Current density capped at  $\leq 0.5 \text{ mA}\cdot\text{cm}^{-2}$  (safe/used in study) to open aqueous pathways and drive cargo through MN microchannels.
- **Stage-2 (systemic→CNS): RVG-functionalized nanoparticles:** **MSN core** (high loading of small molecules), **PLGA shell** (sustained release & protection), **chitosan (CS) outer coat** (positive  $\zeta$  to enhance skin/lymph transit and uptake), and **PEGylation** for colloidal stability. Dual payload: **Favipiravir** (broad-spectrum RdRp inhibitor) and **Ranpirnase** (RNase with potent anti-RABV in vitro). RVG targets nAChR/GABA receptors to cross BBB and accumulate in neural tissue.

API Schematic: RVG-Targeted Dual-Drug Nanoparticle  
(MSN Core • PLGA Shell • Chitosan Coat)



### Rationale from the papers:

- MN+iontophoresis already doubled model-drug spread and elevated dog IgG/RVNA vs MN alone/IM, proving the front-end delivery advantage and skin safety.
- PLGA/CS nanoparticulates show sustained release, low blood/genotoxicity, and tunable surfaces.
- RVG-MSN improved CNS drug levels with **~13.9% loading** and **~31.4% cumulative release at 192 h**, yielding **77% survival** in a neurotropic infection model.
- Ranpirnase failed in vivo at standard dosing (delivery barrier), despite **IC<sub>50</sub> down to 0.2–2 nM**, so targeted CNS delivery is the missing piece.

| Method from literature                                             | What it gives us                                                     | How we used it in the final system                                                            |
|--------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Microneedle patch (MN)</b> <sup>[1]</sup>                       | Painless delivery through skin + vaccine effect                      | Our patch is made of the same HA/PVP microneedles and contains rabies antigen (for immunity)  |
| <b>Iontophoresis</b> <sup>[1]</sup> (weak electric current)        | Speeds up drug entry through microneedle channels                    | We add a small battery layer so drugs move faster into the blood                              |
| <b>PLGA / chitosan nanoparticles</b> <sup>[2]</sup>                | Safe, slow-release, low toxicity particles                           | We use PLGA shell + chitosan coat for long sustained release + better absorption              |
| <b>MSN nanoparticles</b> (mesoporous silica) <sup>[3]</sup>        | Excellent drug loading (esp. Favipiravir)                            | We keep the MSN <i>inside</i> the PLGA shell so drug releases in two phases                   |
| <b>RVG peptide targeting</b> <sup>[3]</sup>                        | Allows particles to cross the blood-brain barrier                    | We attach RVG to the nanoparticle surface so the drug reaches the brain (critical for rabies) |
| <b>Ranpirnase antiviral study</b> <sup>[4]</sup>                   | Shows a great antiviral <i>that failed only due to poor delivery</i> | We solve its failure by loading it in our brain-targeted nanoparticle                         |
| <b>Favipiravir study</b> <sup>[3]</sup>                            | Strong antiviral + proven in RVG-nanoparticles                       | We keep Favipiravir as the first antiviral payload                                            |
| <b>MN + iontophoresis</b> <sup>[1]</sup> <b>animal immune data</b> | Proven to give better antibody levels than injections                | So our patch gives both <i>immunity</i> and <i>therapy</i> in one shot                        |

### 3.2 Key Design Parameters (from literature)

- **MN matrix:** HA (1–4% w/v), PVP (20–40% w/v), PEG-400 (5–15% w/w of polymer), folding endurance ~200 cycles; ~70% swelling before loss of integrity. Needle arrays formed by vacuum micromolding (PDMS molds; 150–500  $\mu\text{m}$  options).
- **Nanoparticles:** Size progression: MSN ~120 nm  $\rightarrow$  T-705 & MSN ~168 nm  $\rightarrow$  **T-705 & MSN-RVG ~238 nm**;  $\zeta$  potential shifts with RVG/CS/PEG. Encapsulation rate ~61.6%, loading ~13.9%, sustained release to 192 h.
- **Safety:** MN histology shows epidermal grooves without deep vascular damage (rats). PLGA/CS formulations showed negligible hemolysis/genotoxicity.

### 3.3 Transport/Release Calculations (order-of-magnitude, literature-anchored)

#### (A) Dermal flux gain with iontophoresis over MN alone.

The study's model dye permeation area was  $\sim 2\times$  with MN+iontophoresis vs MN alone (MNdi vs MNdo). Assuming similar scaling for nanoparticle convective-electroosmotic transport through

MN-created low-resistance channels at  $\leq 0.5 \text{ mA}\cdot\text{cm}^{-2}$ , we apply a conservative **1.8× flux multiplier** (reported for small molecule; also consistent with macromolecule 4.8× in other MN+ionto contexts cited therein). This justifies shorter wear times ( $\leq 30\text{--}60 \text{ min}$ ) to achieve target particle dose to blood.

#### **(B) Favipiravir loading and release from RVG NP.**

From **13.9% loading** and **31.4% release at 192 h**, a **10 mg NP** dose carries **~1.39 mg** Favipiravir, releasing **~0.44 mg** over 8 days ( $\sim 55 \mu\text{g/day}$  average, front-loaded by diffusion). For **20 mg NP**,  $\sim 0.88 \text{ mg}$  released over 8 days; this sustained trickle supplements an initial burst (below) to keep CNS levels above  $\text{EC}_{50}$  windows used in the RVG-MSN model that achieved 77% survival.

#### **(C) Ranpirnase co-encapsulation target.**

Aim brain extracellular levels  $\geq 20\text{--}50 \text{ nM}$  to cover cell-to-cell spread  $\text{IC}_{50}$  range ( $20\text{--}600 \text{ nM}$ ) with safety headroom. If **5% w/w** Ranpirnase relative to Favipiravir load is co-encapsulated (protein load limited by shell), **20 mg NP** would carry  $\sim 69.5 \mu\text{g}$  Favipiravir-equivalent drug and  $\sim 3.5 \mu\text{g}$  Ranpirnase per mg NP; selecting **20 mg NP** total gives **~70  $\mu\text{g}$  Ranpirnase** packaged. With RVG targeting and local release in brain interstitium, modeled micro-domain concentrations can transiently exceed tens of nM despite systemic dilution—precisely the gap in the failed systemic hamster study.

#### **(D) Staged release profile (core–shell).**

Adopt **MSN core (fast) + PLGA shell (slow)**:

- **Burst phase (first 24–48 h):**  $\sim 10\text{--}15\%$  of Favipiravir and 10% of Ranpirnase diffuse through hydrated PLGA pores (Higuchi: ( $Q = k_H * \sqrt{t}$ )) supporting rapid post-exposure suppression peripherally and during early neuroinvasion.
- **Sustain (2–8 days):** approach the observed **31.4% cumulative at 192 h** for T-705, matching the RVG-MSN paper's profile; tune PLGA (50:50) MW and end-groups to set ( $k_H$ ).

#### **(E) Particle size & charge for trafficking.**

**~200–250 nm** RVG-NPs balance lymphatic uptake/systemic circulation and BBB receptor-mediated transcytosis; **CS coat** provides modest positive  $\zeta$  (enhanced epithelial transit), while **PEG** limits opsonization. Sizes and  $\zeta$  evolution are consistent with the MSN→T-705 & MSN→T-705 & MSN-RVG trajectory in the paper; PLGA shell thickness is adjusted to keep hydrodynamic diameter  $\leq 250\text{--}280 \text{ nm}$  post-CS.

### **3.4 Practical Formulation Plan**

1. **MN patch (vaccine & “primer”):** HA 2% + PVP K30 30% + PEG-400 10% (w/w polymer), 300–500  $\mu\text{m}$  needles, backing contains reconstituted inactivated rabies antigen; fabricate by vacuum micromolding; validated folding endurance ( $\sim 200$ ),  $\sim 70\%$

swelling window. Apply with **0.3–0.5 mA·cm<sup>-2</sup>** iontophoresis for 20–30 min. **Outcome:** higher dermal uptake and stronger IgG/RVNA than IM.

2. **Dual-antiviral NP suspension (co-delivered under the same patch):**

**Core:** MSN loaded with Favipiravir (T-705) via solvent impregnation to ~14% loading.

**Shell:** PLGA (50:50) to modulate burst/sustain.

**Surface:** CS (for cationic interface) + PEG; **RVG peptide** covalently conjugated (as in the paper). Target size **~230–260 nm**; verify  $\zeta$  and loading via DLS/HPLC.

3. **Dose concept (preclinical translation, per 10–15 kg dog or 25–30 g mouse scaling):**

a. **Favipiravir:** NP dose **10–20 mg** → **1.4–2.8 mg payload** with **0.44–0.88 mg** released over 8 days.

b. **Ranpirnase:** co-load to achieve **≥20–50 nM** local brain levels post-BBB transit (guided by IC<sub>50</sub> bands). **Note:** exact mg·kg<sup>-1</sup> set after PK pilot due to species-dependent CNS exposure.

4. **Safety & assays:**

a. **In vitro:** hemolysis/genotoxicity per PLGA/CS nanoparticle paper; cell viability with Ranpirnase/Favipiravir combination; release profiling to 192 h.

b. **Ex vivo/animal:** skin histology after MN+iontophoresis (expect epidermal grooves only); serum IgG/RVNA; brain drug levels; survival in neurotropic challenge.

#### 4. Expected Results and Significance

##### Efficacy:

- **Immune response:** Expect ≥MNvi-like boosts (≥2× RVNA threshold) due to identical front-end and antigen delivery.
- **Antiviral effect:** RVG targeting + staged release should elevate CNS exposure for **Favipiravir** and newly **deliver Ranpirnase** where it failed previously, covering both viral replication (RdRp) and intercellular spread (RNase), mirroring **77% survival** seen with RVG-MSN in neurotropic models-with added dual-mechanism robustness.

##### Safety:

- **Dermal:** MN+iontophoresis shown safe to deeper vasculature.
- **Systemic:** PLGA/CS nanoparticles had negligible hemolysis/genotoxicity; adjust Ranpirnase load to avoid off-target RNase cytotoxicity (benefits from local CNS release rather than high systemic dosing).

##### Practical significance:

A single outpatient, patch-based procedure that **primes immunity and immediately starts CNS-targeted antiviral therapy** could simplify logistics, reduce cold-chain needle use, and bridge the critical window between exposure and symptom onset.

## 5. Gaps, Drawbacks, Barriers-and How the Consolidated Design Addresses Them

| Source method                                  | Key gap/drawback                                                 | Consolidated fix                                                                           |
|------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| MN vaccination + iontophoresis <sup>[41]</sup> | No CNS drug delivery                                             | Add RVG-NP antivirals under the same patch; iontophoresis drives NPs in.                   |
| PLGA/CS antigen NPs <sup>[21]</sup>            | Immune focus; no BBB targeting; vaccine only                     | Keep sustained-release virtues but add RVG and antiviral co-loads for therapy.             |
| RVG-MSN (Favipiravir) <sup>[31]</sup>          | Single antiviral; relies on high dose; no front-end acceleration | Co-deliver Ranpirnase + MN-ionto front-end for faster entry and dual mechanism.            |
| Ranpirnase systemic <sup>[41]</sup>            | No BBB transit; failed in vivo                                   | Encapsulate and RVG-target to brain; stage release to hit IC <sub>50</sub> levels locally. |

### Foreseeable barriers & mitigations

- **Protein stability (Ranpirnase) inside PLGA:** Use inner **MSN adsorption + PLGA shell** with end-capped 50:50 and pH-neutral microenvironment; add trehalose/albumin excipients (albumin also explored as surface modifier previously).
- **Ligand density tuning:** Optimize RVG valency to balance targeting vs RES clearance (measure  $\zeta$  and binding uptake).
- **Iontophoresis comfort & standardization:** Stay  $\leq 0.5 \text{ mA}\cdot\text{cm}^{-2}$ ; short sessions validated in study.

## 6. Discussion

### Stability analysis

- a. Microneedle front-end (HA/PVP/PEG-400).
  - The MNs used in the rabies paper withstand ~200 folds, show ~70% swelling in ~3.5 min, and keep sharp, smooth microprojections-evidence of good mechanical/handling stability and rapid hydration on skin.
  - Iontophoresis at  $0.5 \text{ mA}\cdot\text{cm}^{-2}$  was used and deemed safe; microchannels close within ~48 h, so barrier recovery is expected without deep vascular injury.
- b. Nanoparticle cargo (MSN & PLGA-CS/PEG with RVG).
  - Favipiravir loading ~13.9% (w/w) with encapsulation 61.6%; particles enlarge after payload + RVG conjugation (TEM), consistent with stable surface modification.
  - In-vitro cumulative release 31.4% at 192 h (8 days), indicating diffusion-controlled sustained release that we can further slow with PLGA shell thickness/MW.
  - Protein (Ranpirnase) stability risk inside polyesters is mitigated by (i) adsorption within MSN first (less acid exposure), (ii) end-capped PLGA 50:50, and (iii) stabilizers (trehalose/albumin)-all common NP practice and compatible with the PLGA/CS antigen paper's benign chemistry.



## Permeability analysis (layer-wise)

Barrier model (MN patch + iontophoretic driving):

- Stratum corneum (SC): MNs bypass SC via 300–500  $\mu\text{m}$  microchannels; histology shows epidermal grooves only, no deep vessel damage. Net effect: SC resistance  $\approx$  removed in channel area.
- Viable epidermis/dermis: After SC bypass, the limiting steps are intercellular/appendageal pathways and capillary uptake. With iontophoresis ( $0.5 \text{ mA}\cdot\text{cm}^{-2}$ ), permeation area/flux  $\approx 2\times$  vs MN alone for a model drug; literature within the same paper cites up to  $4.8\times$  increases for macromolecules with MN+ionto. We conservatively use  $\sim 1.8\text{--}2.0\times$  flux gain for our NP suspension through MN-created low-resistance channels.
- Lymphatics/systemic: Once past dermis, nanoparticles ( $\approx 230\text{--}260 \text{ nm}$  hydrodynamic diameter target) drain to lymph and blood. Positive CS coat helps epithelial transit/uptake; PEG improves colloidal stability.
- BBB crossing: RVG (29-aa) on the surface supports receptor-mediated transport into brain; the RVG–MSN system reduced brain viral load and improved survival to 77% at 21 dpi in a neurotropic model-functional evidence of BBB entry.

## Cytotoxicity / biocompatibility

- PLGA-b-PEG and CS-PEG NPs with rabies antigen showed no significant hemolysis and no DNA fragmentation (genotoxicity) in human blood assays-good systemic compatibility signals for our polymer choices and CS surface.
- The RVG–MSN–Favipiravir system decreased virus titers without inducing substantial cell damage in vitro, supporting acceptable intrinsic cytocompatibility at effective doses.
- Iontophoresis safety at  $0.5 \text{ mA}\cdot\text{cm}^{-2}$  is explicitly discussed as acceptable; MN + ionto did not damage deep vasculature in rat skin.

## Pharmacokinetics (PK) – what we expect

Compartment chain: Skin depot (MN channels)  $\rightarrow$  lymph/blood (ionto-assisted)  $\rightarrow$  systemic NP distribution (RES + circulation)  $\rightarrow$  RVG-mediated BBB influx  $\rightarrow$  brain interstitium depot (diffusion-limited release).

Key anchors from the literature:

- Drug loading & release: Favipiravir 13.9% loading, 31.4% released by 192 h from RVG–MSN, implying a sustained input term to plasma/brain for  $\sim 1$  week+ post-dose.
- CNS delivery efficacy: RVG–MSN–Favipiravir improved survival to 77% vs 23% control in a neurotropic mouse model-indirect PK/PD evidence of brain exposure exceeding antiviral threshold.
- Ranpirnase systemic limits: Free Ranpirnase showed in-vivo failure at  $0.1 \text{ mg}\cdot\text{kg}^{-1}$  and short serum  $t_{1/2}$  ( $\sim 36.5 \text{ min}$ ), implying that formulation and BBB targeting are necessary to raise brain exposure/time above its  $\text{IC}_{50}$  bands.

### Design targets (PK parameters to estimate in first pilots):

- NP size  $\approx$  230–260 nm post-coatings to balance circulation vs BBB transcytosis (measured by DLS).
- Brain:plasma  $K_p$ , brain (apparent) should rise above small-molecule control (Favipiravir alone) thanks to RVG; verify by tissue AUC ratios while tracking RES clearance. (Evidence for CNS delivery is functional via survival in the RVG paper.)
- Dermal input ( $J_{in}$ ): assume MN+ionto  $\sim 2\times$  flux vs MN alone during the 20–30 min wear time, defining the initial systemic NP bolus.

### Pharmacodynamics (PD) – combined antiviral & immune effects

- Favipiravir blocks viral RdRp and, when targeted with RVG–MSN, reduces brain viral RNA and improves survival (77% at 21 dpi), indicating that CNS concentrations achieved are pharmacodynamically meaningful.
- Ranpirinase (RNase) inhibits RABV release at 0.2–2 nM and cell-to-cell spread at 20–600 nM in vitro. Our targeting + sustained release aim to lift local brain interstitial levels into these bands long enough to suppress spread-precisely where free drug failed in vivo.
- Immune PD: MN+ionto vaccination raises IgG ( $\sim 206\%$  vs control) and RVNA ( $\sim 2.2\times$  threshold) in dogs, creating concurrent neutralizing antibody support while antivirals act-two levers working in parallel.

### Expected release rate

Using the RVG–MSN Favipiravir data (same chemistry we embed inside PLGA/CS to further tune):

- Per-particle kinetics (anchor): 31.4% cumulative at 192 h from T-705 & MSN-RVG. If you administer 20 mg NP, at 13.9% loading that's 2.78 mg Favipiravir encapsulated; 31.4% of 2.78 mg  $\approx$  0.872 mg released over 192 h.  
→ Average release rate  $\approx$  0.872 mg / 192 h  $\approx$  4.54  $\mu\text{g}\cdot\text{h}^{-1}$   $\approx$  109  $\mu\text{g}\cdot\text{day}^{-1}$ .  
Expect a front-loaded diffusion phase (Higuchi-like) followed by slower release; a thin PLGA shell produces a modest initial burst (10–15%) then sustains (you can reduce burst by thickening/raising PLGA MW).
- Ranpirinase: No direct release curve in the packets; we set a design target of  $\leq 10\%$  burst in 24 h and  $\sim 30\text{--}50\%$  over 7–10 days by (i) adsorbing to MSN (less acid), (ii) adding a PLGA shell and CS/PEG outer layer, and (iii) using protein stabilizers. First in-vitro release in artificial CSF (pH 7.3) + protease guard will tune this to keep brain interstitial levels in the 20–600 nM “spread-blocking” window. (Rationale grounded in Ranpirinase  $\text{IC}_{50}$  bands and the Favipiravir MSN release profile.)

## 7. Conclusion

By merging **MN-iontophoresis vaccination**, **PLGA/CS sustained release**, **RVG BBB-targeting**, and **dual-antiviral loading (Favipiravir + Ranpirnase)** into a **single patch procedure**, the proposed platform directly addresses the chief barriers-**skin barrier**, **BBB**, and **monotherapy limitations**-to produce stronger immune priming and therapeutic CNS drug levels with a favorable safety profile.

## 8. References

1. [MN-iontophoresis microneedles \(Pharm Res. 2023\)](#) ;
2. [PLGA/CS antigen nanoparticles \(Resource-Efficient Technologies, 2016\)](#) ;
3. [RVG-Favipiravir MSN nanoparticles \(IJMS, 2023\)](#) ;
4. [Ranpirnase antiviral against rabies \(Viruses, 2020\)](#)

